

Permutation Tests for 2 Samples - Lab Activity - Example Solution

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In this lab, you will get to try a few examples of permutation tests.

Example 1 is in the examples file for your reference.

Example 2 - Machine Repair

An automatic refill machine (16 oz expected) just underwent repairs. Here are the reported oz for sampled beverages before and after the repairs.

```
before <- c(16.55, 15.36, 15.94, 16.43, 16.01)
after  <- c(16.05, 15.98, 16.10, 15.88, 15.91)
```

Run a `perm.test` command to test whether or not there was a shift in means between the before/after repair amounts of liquid put into 16 oz beverages. Obtain a 90 percent CI for the difference in means.

SOLUTION:

```
perm.test(before, after, conf.int = TRUE, conf.level = 0.90)

##
## 2-sample Permutation Test (scores mapped into 1:(m+n) using rounded
## scores)
##
## data:  before and after
## T = 29, p-value = 0.9365
## alternative hypothesis: true mu is not equal to 0
## 90 percent confidence interval:
## -3.0 3.5
```

Other methods, including the Wilcoxon Rank Sum procedure, do not find enough evidence to conclude that there is a difference in centers here. Do you reach the same conclusion here?

SOLUTION:

Yes. The p-value is very large, and the 90% confidence interval includes zero. Thus, we do not have evidence of a shift in the means.

How many possible permutations were there for this data set?

SOLUTION:

```
m <- 5
n <- 5
possperm <- choose(m+n, m); possperm
```

```
## [1] 252
```

252 permutations

Here is what a simulated distribution looks like.

```

set.seed(225)
mean(before)-mean(after)

## [1] 0.074

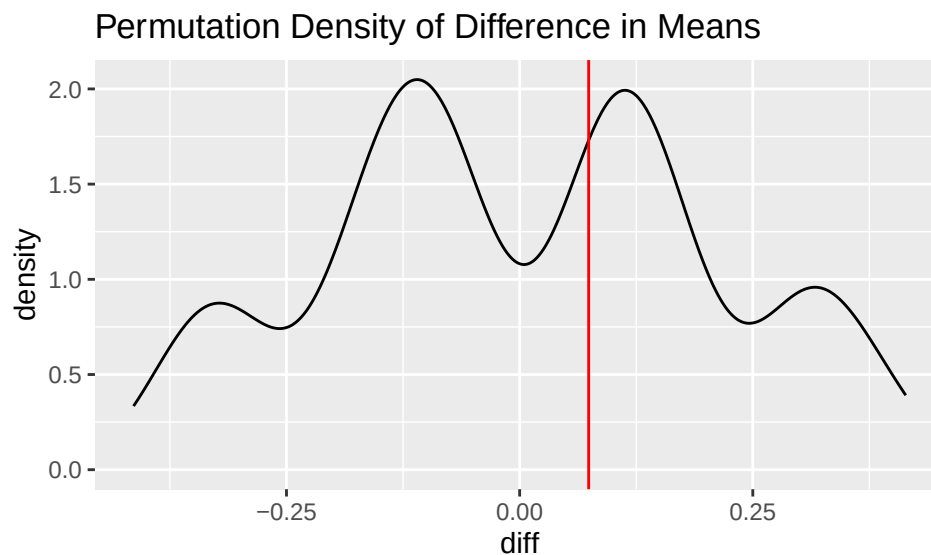
runs <- 1000
diff <- rep(0,runs)
sampledata <- c(before, after)
total <- sum(sampledata)

for(i in 1:runs){
  sampley <- sample(sampledata, 5)
  sumy <- sum(sampley)
  diff[i] <- (total-sumy)/5-sumy/5
}

df <- data.frame(diff)

ggplot(data = df, aes(x = diff)) +
  geom_density() +
  geom_vline(xintercept = 0.074, color = "red") +
  labs(title = "Permutation Density of Difference in Means")

```



Our observed value is very near the middle, which is why we did not have evidence to conclude there was a shift in the distributions. We should obtain the same result here.

Again, you can get empirical p-values (meaning estimated in this way) using:

```

2*pnorm(diff, 0.074, lower.tail = FALSE, data = df)

## [1] 0.842

# what alpha does this correspond to?
qdata(~ diff, c(0.05, 0.95), data = df) # 5% and 95% bounds

##      5%      95%
## -0.338  0.346

```

You would reject the null hypothesis if you had observed a sample difference in means that fell outside the provided bounds. Where does 0.074 fall? Does this confirm that you do not reject the null hypothesis?

SOLUTION:

0.074 is well within (-0.338, 0.346), so this confirms that we would not reject the null hypothesis.

Example 3 - Iris Data

The iris data set is a fairly famous data set. You can read about it if you ask for help about the data set.

```
data(iris)
# help(iris)
names(iris)
```

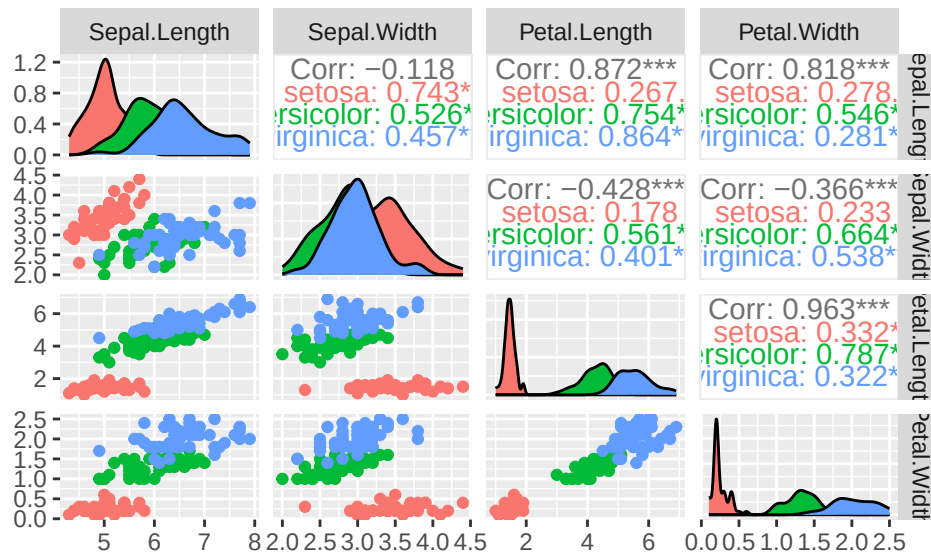
```
## [1] "Sepal.Length" "Sepal.Width" "Petal.Length" "Petal.Width" "Species"
```

The data set contains measurements of 4 different variables for three different species of Iris, 50 observations per species. Here, we'll break down the data set into one per species, to help with running the perm.test function later.

```
setosa <- filter(iris, Species == "setosa")
versicolor <- filter(iris, Species == "versicolor")
virginica <- filter(iris, Species == "virginica")
```

A quick look at the data set reveals some species may have similar traits and others not.

```
ggpairs(iris, columns = 1:4, aes(color = Species))
```



Let's examine a few different questions about this data set using permutation tests.

Do Versicolor and Virginica have different mean Sepal widths?

How many possible permutations would have to be listed to cover them all?

SOLUTION:

```
m <- 50
n <- 50
possperm <- choose(m+n, m); possperm
```

```
## [1] 1.008913e+29
```

That's a lot of permutations...

Run a permutation test to address the question of interest.

```
perm.test(versicolor$Sepal.Width, virginica$Sepal.Width)
```

```
##
## 2-sample Permutation Test (scores mapped into 1:(m+n) using rounded
## scores)
##
## data: versicolor$Sepal.Width and virginica$Sepal.Width
## T = 2141, p-value = 0.00185
## alternative hypothesis: true mu is not equal to 0
```

What do you conclude?

SOLUTION:

There is significant evidence of a difference in mean Sepal Widths (empirical p-value is 0.00185).

Here is what the distribution of simulated differences in means would look like:

```
mean(versicolor$Sepal.Width)-mean(virginica$Sepal.Width)
```

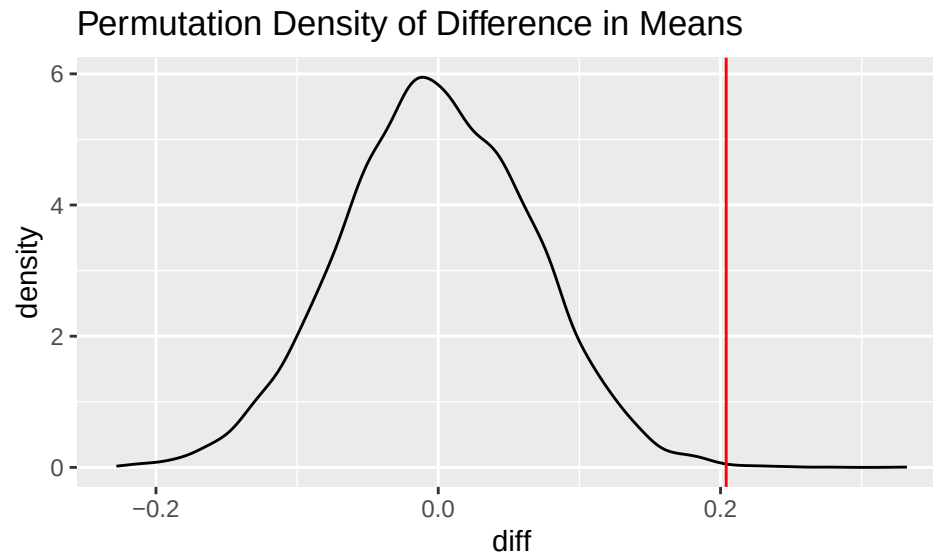
```
## [1] -0.204
```

```
set.seed(225)
runs <- 10000
diff <- rep(0, runs)
sampledata <- c(versicolor$Sepal.Width, virginica$Sepal.Width)
total <- sum(sampledata)

for(i in 1:runs){
  sampley <- sample(sampledata, 50)
  sumy <- sum(sampley)
  diff[i] <- (total-sumy)/50-sumy/50
}

df <- data.frame(diff)

ggplot(data = df, aes(x = diff)) +
  geom_density() +
  geom_vline(xintercept = 0.204, color = "red") +
  labs(title = "Permutation Density of Difference in Means")
```



Obtain an empirical p-value for your test procedure.

SOLUTION:

```
mydata <- data.frame(diff) # insists on the values being in a dataframe
# assume symmetry and just do lower tail
2*pdata(mydata$diff, -0.204)
```

```
## [1] 0.0022
```

```
# get each tail in actuality from simulation
pdata(mydata$diff, -0.204) + 1-pdata(mydata$diff, 0.204)
```

```
## [1] 0.0023
```

Obtain an empirical 0.05 lower-tail cutoff for a rejection region based on the simulation.

SOLUTION:

```
qdata(mydata$diff, 0.05)
```

```
##      5%
## -0.112
```

Do your findings confirm rejecting the null hypothesis?

SOLUTION:

Yes. Though the RR is set up for a higher alpha (0.10, with 0.05 in both tails), we see our -0.204 value is less than that cutoff and the empirical two-sided p-value is tiny.

Use permutation procedures to address the following questions:

How much difference is there between the mean of setosa sepal length compared to versicolor's sepal length?

(Think: Is a test or CI more useful to address this question?)

SOLUTION:

```
perm.test(setosa$Sepal.Length, versicolor$Sepal.Length, conf.int = TRUE)
```

```
##
## 2-sample Permutation Test (scores mapped into 1:(m+n) using rounded
## scores)
##
## data: setosa$Sepal.Length and versicolor$Sepal.Length
## T = 1313, p-value < 2.2e-16
## alternative hypothesis: true mu is not equal to 0
## 95 percent confidence interval:
## -40.82609 -27.81481
```

We want a CI. We see a 95% CI of (-40.83, -27.81) for our first run. (Your confidence limits should be similar, but they will not be exactly the same as the confidence limits we simulated.) Versicolor's mean sepal length seems to be larger by between 27.81 and 40.83 units (We know this seems large, but perm.test uses a transformed set of scores, so be careful with the units.) compared to setosa's mean sepal length.

Is there significant evidence that the versicolor mean petal length is different from that of virginica's?

SOLUTION:

```
perm.test(versicolor$Petal.Length, virginica$Petal.Length)
```

```
##
## 2-sample Permutation Test (scores mapped into 1:(m+n) using rounded
## scores)
##
## data: versicolor$Petal.Length and virginica$Petal.Length
## T = 1617, p-value < 2.2e-16
## alternative hypothesis: true mu is not equal to 0
```

Yes. The p-value is tiny.

Is there a significant difference in the mean of Setosa sepal width and mean of virginica sepal width using only a subset of the observations as the data source? Obtain an exact p-value/conclusion by hand.

Here are the first 3 setosa observations, and the first 2 virginica observations. Note: there are only TEN possible permutations. Each permutation is equally likely to occur. (You may want scratch paper.)

```
setosa[1:3, ]
```

```
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
## 1          5.1         3.5         1.4         0.2  setosa
## 2          4.9         3.0         1.4         0.2  setosa
## 3          4.7         3.2         1.3         0.2  setosa
```

```
virginica[1:2, ]
```

```
##   Sepal.Length Sepal.Width Petal.Length Petal.Width  Species
## 1          6.3         3.3         6.0         2.5 virginica
## 2          5.8         2.7         5.1         1.9 virginica
```

```
choose(5, 3)
```

```
## [1] 10
```

SOLUTION:

The idea here is to list all 10 ways to get 2 obs in for Virginica. You get the ten different possible differences in sample means, and look to see where your observed value falls. We found the means of the y's (Virginica), used that to find the means of the x's (Setosa), and then found the difference between them.

```
ybar <- c(2.85, 2.95, 3.0, 3.1, 3.1, 3.15, 3.25, 3.25, 3.35, 3.4)
xbar <- (15.7-2*ybar)/3
diff <- xbar-ybar
sort(diff)
```

```
## [1] -0.43333333 -0.35000000 -0.18333333 -0.18333333 -0.01666667  0.06666667
## [7]  0.06666667  0.23333333  0.31666667  0.48333333
```

Our observed difference was 0.23. In the differences vector, there are 3 values from here up, and that's only one sided, so the p-value is at least 0.3. We would not reject the null based on this small sample size.

Is there a significant difference in the mean of setosa sepal width and mean of virginica sepal width using all observations from both species? Do you make a different conclusion than above?

SOLUTION:

```
perm.test(setosa$Sepal.Width, virginica$Sepal.Width)
```

```
##
## 2-sample Permutation Test (scores mapped into 1:(m+n) using rounded
## scores)
##
## data: setosa$Sepal.Width and virginica$Sepal.Width
## T = 2792, p-value = 3.338e-09
## alternative hypothesis: true mu is not equal to 0
```

We reject the null hypothesis here. There is sufficient evidence that there is a significant difference in mean sepal width between setosa and virginica. This is a different result than we obtained when only using a subset of the values.

Data Sources

Machine Repair data was simulated.

Iris data. Available in R with data(iris). Original sources referenced in R help:

Fisher, R. A. (1936) The use of multiple measurements in taxonomic problems. *Annals of Eugenics*, 7, Part II, 179–188. doi:10.1111/j.1469-1809.1936.tb02137.x.

Anderson, Edgar (1935). The irises of the Gaspé Peninsula, *Bulletin of the American Iris Society*, 59, 2–5.

Standard Reference

These materials were created for use in an undergraduate nonparametrics course. Where provided, textbook references refer to *Nonparametric Statistical Methods* (3rd Edition) by Hollander, Wolfe, and Chicken. Notation is chosen to be consistent with that text. However, materials may be used independently of the textbook.

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